2.6.28 :

$$\frac{x+3}{x-2}+\frac{x+2}{x-3}=\frac{x+3}{x-3}-\frac{x+2}{x-2} \qquad /\cdot(x-2)(x-3)$$

$$(x+3)(x-3)+(x+2)(x-2)=(x+3)(x-2)-(x+2)(x-3) \qquad \mathcal{D}: \boxed{x\neq 2\wedge x\neq 3}$$

$$x^2-9+x^2-4=x^2+x-6-x^2+x+6$$

$$2x^2-2x-13=0$$

$$D=(-2)^2-4\cdot 2\cdot(-13)=4+104=108 \qquad \sqrt{D}=6\sqrt{3}$$

$$x_{1,2}=\frac{2\pm 6\sqrt{3}}{4}=\left\langle\begin{array}{l}\frac{2+6\sqrt{3}}{4}=\frac{1}{2}+\frac{3}{2}\sqrt{3}\in\mathcal{D}\\ \frac{2-6\sqrt{3}}{4}=\frac{1}{2}-\frac{3}{2}\sqrt{3}\in\mathcal{D}\end{array}\right.$$

$$\mathbf{K=\{\tfrac{1}{2}\pm\tfrac{3}{2}\sqrt{3}\}}$$

2.6.29 :

$$\frac{1}{x-4}-\frac{1}{x+4}=\frac{1}{x-5}-\frac{1}{x-2} \qquad /\cdot(x-4)(x+4)(x-5)(x-2)$$

$$(x+4)(x-5)(x-2)-(x-4)(x-5)(x-2)=(x-4)(x+4)(x-2)-(x-4)(x+4)(x-5) \qquad \mathcal{D}: \boxed{x\in\mathcal{R}-\{4;-4;5;2\}}$$

$$(x+4)(x^2-7x+10)-(x-4)(x^2-7x+10)=(x^2-16)(x-2)-(x^2-16)(x-5)$$

$$x^3-3x^2-18x+40-x^3+11x^2-38x+40=x^3-2x^2-16x+32-x^3+5x^2+16x-80$$

$$8x^2-56x+80=3x^2-48$$

$$5x^2-56x+128=0$$

$$D=(-56)^2-4\cdot 5\cdot 128=3136-2560=576 \qquad \sqrt{D}=24$$

$$x_{1,2}=\frac{56\pm 24}{10}=\left\langle\begin{array}{l}\frac{80}{10}=8\in\mathcal{D}\\ \frac{32}{10}=\frac{16}{5}\in\mathcal{D}\end{array}\right.$$

$$\mathbf{K=\{8;\tfrac{16}{5}\}}$$

2.6.30 :

$$\frac{1}{x-5}-\frac{1}{x+3}=\frac{1}{x-6}-\frac{1}{x-3} \qquad /\cdot(x-5)(x+3)(x-6)(x-3)$$

$$(x+3)(x-6)(x-3)-(x-5)(x-6)(x-3)=(x-5)(x+3)(x-3)-(x-5)(x+3)(x-6) \qquad \mathcal{D}: \boxed{x\in\mathcal{R}-\{5;-3;6;3\}}$$

$$(x+3)(x^2-9x+18)-(x-5)(x^2-9x+18)=(x^2-2x-15)(x-3)-(x^2-2x-15)(x-6)$$

$$x^3-6x^2-9x+54-x^3+14x^2-63x+90=x^3-5x^2-9x+45-x^3+8x^2+3x-90$$

$$8x^2-72x+144=3x^2-6x-45$$

$$5x^2-66x+189=0$$

$$D=(-66)^2-4\cdot 5\cdot 189=4356-3780=576 \qquad \sqrt{D}=24$$

$$x_{1,2}=\frac{66\pm 24}{10}=\left\langle\begin{array}{l}\frac{90}{10}=9\in\mathcal{D}\\ \frac{42}{10}=\frac{21}{5}\in\mathcal{D}\end{array}\right.$$

$$\mathbf{K=\{9;\tfrac{21}{5}\}}$$

2.6.31 :

$$\frac{\frac{x}{3}-\frac{x-1}{2}}{\frac{x}{3}-\frac{x+1}{4}}=x \qquad \mathcal{D}: \boxed{x\neq 3}$$

$$\frac{\frac{2x-3x+3}{6}}{\frac{4x-3x-3}{12}}=x$$

$$\frac{-(x-3)}{6}\cdot\frac{12}{x-3}=x$$

$$x=-2\in\mathcal{D}$$

$$\mathbf{K=\{-2\}}$$

Příklady k procvičení

Řešte v $\mathcal{R}$:

2.6.P.1 : $x-\frac{5}{x}=\frac{11}{4}$ — $K=\{4;-\frac{5}{4}\};\mathcal{D}: \boxed{x\neq 0}$

2.6.P.2 : $\frac{3x}{x+6}-\frac{4}{x-1}=4$ — $K=\{0;-27\};\mathcal{D}: \boxed{x\neq -6\wedge x\neq 1}$

2.6.P.3 : $\frac{4x+5}{x}-\frac{12}{x-2}=1$ — $K=\{5;-\frac{2}{3}\};\mathcal{D}: \boxed{x\neq 0\wedge x\neq 2}$

2.6.P.4 : $\frac{5-3x}{3-5x}+\frac{3-5x}{5-3x}=\frac{5}{2}$ — $K=\{7;\frac{1}{7}\};\mathcal{D}: \boxed{x\neq\frac{3}{5}\wedge x\neq\frac{5}{3}}$

2.6.P.5 : $\frac{1}{x-1}+\frac{1}{x+1}=\frac{9}{4x}$ — $K=\{\pm 3\};\mathcal{D}: \boxed{x\neq\pm 1\wedge x\neq 0}$

2.6.P.6 : $\frac{1}{x}+\frac{4}{x+3}+\frac{4}{x-3}=0$ — $K=\{\pm 1\};\mathcal{D}: \boxed{x\neq\pm 3\wedge x\neq 0}$

2.6.P.7 : $\frac{5}{3-2x}+\frac{4}{x-2}=\frac{5}{x+1}$ — $K=\{4;\frac{8}{7}\};\mathcal{D}: \boxed{x\in\mathcal{R}-\{\frac{3}{2};2;-1\}}$

2.6.P.8 : $\frac{2x+1}{x+3}-\frac{x-1}{x^2-9}=\frac{x+3}{3-x}-\frac{4+x}{3+x}$ — $K=\{1;-\frac{5}{4}\};\mathcal{D}: \boxed{x\neq\pm 3}$

2.6.P.9 : $\frac{1}{x-1}-\frac{3}{x^2+x+1}=\frac{7}{x^3-1}$ — $K=\{3;-1\};\mathcal{D}: \boxed{x\neq -1}$

2.6.P.10 : $\frac{2x-3}{x^3+1}=\frac{1}{x^2-x+1}-\frac{2}{x^2+2x+1}$ — $K=\{1;-\frac{1}{3}\};\mathcal{D}: \boxed{x\neq -1}$

2.6.P.11 : $\frac{1}{x-1}-\frac{1}{x-4}=\frac{1}{x+5}-\frac{1}{x-3}$ — $K=\{7;\frac{11}{5}\};\mathcal{D}: \boxed{x\in\mathcal{R}-\{1;4;-5;3\}}$

2.6.P.12 : $\frac{x-1}{x+1}-\frac{x-2}{x+2}=\frac{x-3}{x+3}-\frac{x-4}{x+4}$ — $K=\{0;-\frac{5}{2}\};\mathcal{D}: \boxed{x\in\mathcal{R}-\{-1;-2;-3;-4\}}$

Řešené příklady

Řešte v $\mathcal{R}$:

2.7.1 : Sestavte kvadratickou rovnici o kořenech $x_1=3;\ x_2=6$

$$V.v.\Rightarrow x_1+x_2=-b\ \wedge\ x_1\cdot x_2=c$$

$$3+6=-b \qquad 3\cdot 6=c$$

$$b=-9 \qquad c=18$$

$$\Rightarrow \quad \mathbf{x^2-9x+18=0}$$

2.7.2 : Sestavte kvadratickou rovnici o kořenech $x_1=-5;\ x_2=3$

$$V.v.\Rightarrow x_1+x_2=-b\ \wedge\ x_1\cdot x_2=c$$

$$-5+3=-b \qquad (-5)\cdot 3=c$$

$$b=2 \qquad c=-15$$

$$\Rightarrow \quad \mathbf{x^2+2x-15=0}$$

2.7.3 : Sestavte kvadratickou rovnici o kořenech $x_1=3;\ x_2=-\frac{1}{2}$

$$V.v.\Rightarrow x_1+x_2=-b\ \wedge\ x_1\cdot x_2=c$$

$$3-\frac{1}{2}=-b \qquad 3\cdot(-\frac{1}{2})=c$$

$$b=-\frac{5}{2} \qquad c=-\frac{3}{2}$$

$$\Rightarrow \quad x^2-\frac{5}{2}x-\frac{3}{2}=0 \quad /\cdot 2$$

$$\mathbf{2x^2-5x-3=0}$$

2.7.4 : Sestavte kvadratickou rovnici o kořenech $x_1=-2;\ x_2=\frac{5}{3}$

$$V.v.\Rightarrow x_1+x_2=-b\ \wedge\ x_1\cdot x_2=c$$

$$-2+\frac{5}{3}=-b \qquad (-2)\cdot\frac{5}{3}=c$$

$$b=-\frac{1}{3} \qquad c=-\frac{10}{3}$$